9.3 Resistivity

Question Paper

Course	CIE A Level Physics
Section	9. Electricity
Topic	9.3 Resistivity
Difficulty	Easy

Time allowed: 40

Score: /26

Percentage: /100

Question 1a

A straight wire of unstretched length L has an electrical resistance R. The area of the cross-section A of the wire may be assumed to be constant.

State the relation between R, L, A and the resistivity ρ of the material of the wire.

[1 mark]

Question 1b

Measurements made for a sample of metal wire are shown in Table 1.1.

Table 1.1

Quantity	Measurement
resistance	6.1 Ω
diameter	0.44 mm
length	1390 mm

State the most appropriate instrument to use to measure

(i)
the resistance of the wire

[1]

(ii)
the diameter of the wire

[1]

(iii)
the length of the wire.

[1]

[3 marks]

Question 1c

Show that the resistivity of the metal is $6.67 \times 10^{-7}\ \Omega\,\text{m}$.

[1 mark]

Question 2a

On Fig 1.1, sketch the change in resistance with temperature for a thermistor.

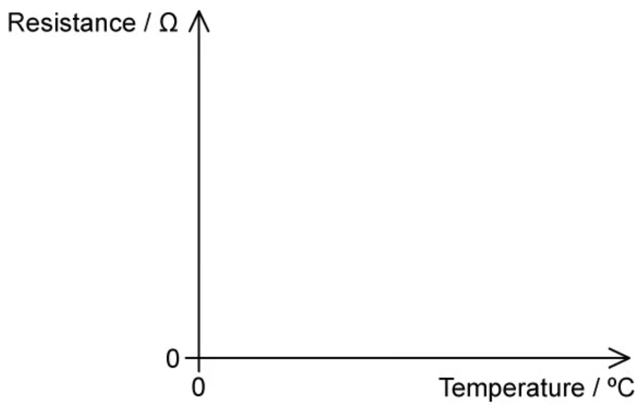


Fig 1.1

[2 marks]

Question 2b

Cobalt is commonly used to create thermistors. At 20 °C a wire of cobalt is used to create a thermistor and has an electrical resistivity of 5.6×10^{-8} Ω m.

State and explain what would happen to the resistivity of the thermistor if its temperature is increased to 60 °C.

[3 marks]

Question 2c

(i)

State the meaning of *LDR*.

[1]

(ii)

On Fig 1.2, sketch the change in resistance with light intensity for an LDR.

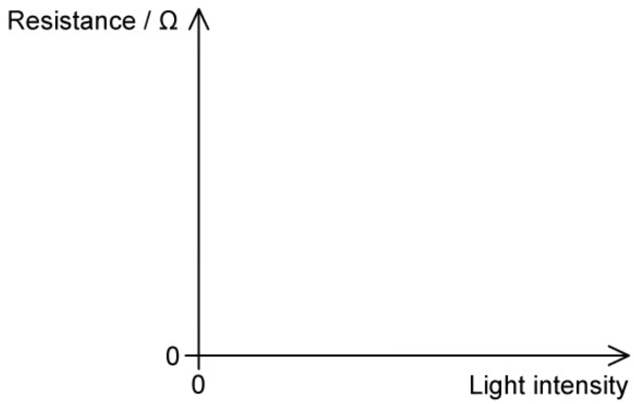


Fig 1.2

[2]

[3 marks]

Question 2d

An LDR is connected to a circuit to operator an automatic street light. The resistance of the LDR is measured to be 5 kΩ at one time and 20 MΩ at another.

State which resistance the LDR is likely to be at during the day and which it will be at night.

[2 marks]

Question 3a

Complete Table 1.1 by placing a tick (✓) in the correct column to show whether each of the quantities affects or does not affect the resistance of a conducting wire.

Table 1.1

quantity	does not affect resistance	affects resistance
cross-sectional area		
length		
resistivity		
mass		

[4 marks]

Question 3b

A lamp is designed to take a current of 0.65 kA when connected across a 240 V mains supply.

The filament is made from one of the three metals from Table 1.2 with a diameter of 66 μm.

Table 1.2

Metal	Resistivity / Ω m
Nichrome	1.1×10^{-6}
Tungsten	5.5×10^{-8}
Copper	1.7×10^{-8}

Calculate the cross-sectional area of the metal wire.

[2 marks]

Question 3c

Calculate the resistance of the wire.

[2 marks]

Question 3d

Hence, or otherwise, determine which metal the filament is made from for a length of 23 mm.

[3 marks]

Model Answers: Easy

1a

The relation between *R, L, A* and the resistivity *ρ* of the material of the wire is:

- $R = \frac{\rho L}{A}$ [1 mark]

[Total: 1 mark]

1b

The most appropriate instrument to use to measure...

(i) The resistance of the wire:

- Voltmeter and ammeter / ohmmeter / multimeter on 'ohm' setting; [1 mark]

(ii) The diameter of the wire:

- Micrometer (screw gauge) / digital / vernier callipers; [1 mark]

(iii) The length of the wire:

- A metre rule / tape measure; [1 mark]

[Total: 3 marks]

1c

Show that the resistivity of the metal is 6.67×10^{-7} Ω m

List the known quantities:

- Resistance, $R = 6.1\ \Omega$
- Diameter, $d = 0.44$ mm $= 0.44 \times 10^{-3}$ m
- Length, $l = 1390$ mm $= 1.39$ m

Determine the equation for the resistivity, ρ:

- $R = \frac{\rho l}{A} \Rightarrow \rho = \frac{RA}{l}$ [1 mark]

Calculate the cross-sectional area, A:

- $A = \pi r^2 = \pi\left(\frac{d}{2}\right)^2 = \frac{\pi d^2}{4}$
- $A = \frac{\pi(0.44 \times 10^{-3})^2}{4} = 1.52 \times 10^{-7}\ \text{m}^2$ [1 mark]

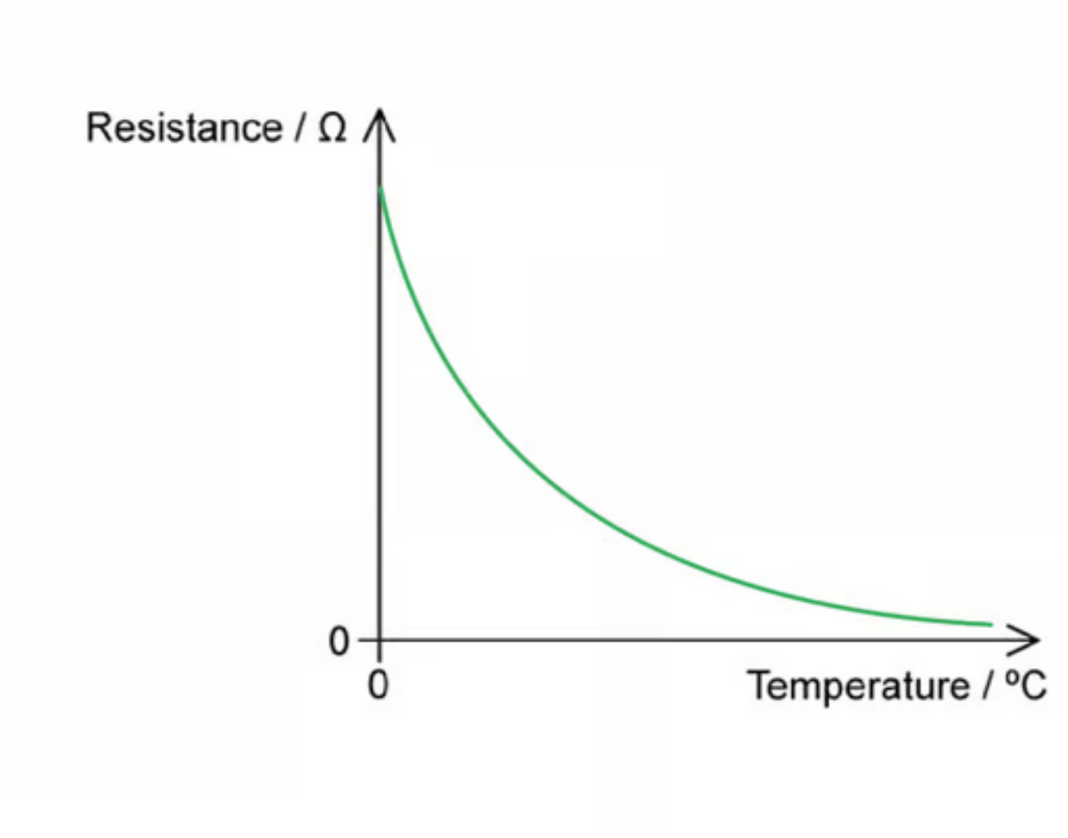
Substitute in the values:

- $\rho = \frac{6.1 \times (1.52 \times 10^{-7})}{1.39}$ [1 mark]
- $\rho = 6.67 \times 10^{-7}\ \Omega\ \text{m}$ [1 mark]

[Total: 4 marks]

2a

A graph of the change in resistance with temperature for a thermistor should look like:



- Curved line showing decreasing gradient with rise in temperature; [1 mark]
- Smooth line not touching temperature axis **AND** no horizontal or vertical line anywhere; [1 mark]

[Total: 2 marks]

2b

At 60 °C, the resistivity of the thermistor would:

- Decrease; [1 mark]

Because:

- The resistance would have decreased; [1 mark]
- Since resistivity is proportional to the resistance / $\rho \propto R$; [1 mark]

[Total: 3 marks]

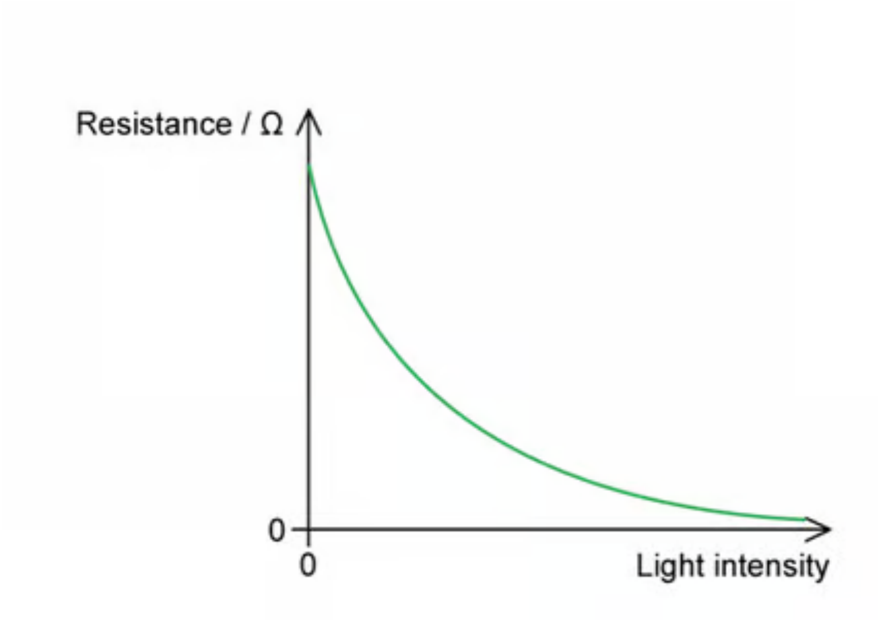
This question links directly back to part (a) of how temperature affects the resistance of a metal. Since the temperature increases, the resistance decreases in the thermistor. From the equation $\rho = \frac{RA}{l}$, where *A* and *l* are still constant, we can see that resistivity is directly proportional to the resistance and hence also decreases.

2c

(i) The meaning of *LDR* is:

- Light-dependent resistor; [1 mark]

(ii) A graph of the change in resistance with light intensity for an LDR should look like:



- Curved line showing decreasing gradient with greater light intensity; [1 mark]
- Smooth line not touching light intensity axis **AND** no horizontal or vertical line anywhere; [1 mark]

[Total: 3 marks]

2d

The resistance the LDR is likely to be at:

- 5 kΩ during the day; [1 mark]
- 20 MΩ during the night; [1 mark]

[Total: 2 marks]

This question links to the graph in part (c).

During the day, the light intensity is going to be very high on the LDR. Therefore, it has a smaller resistance - 5 kΩ. This is the opposite at night.

3a

A table that would score 4 marks should look like:

quantity	**does not affect resistance**	**affects resistance**

3a

A table that would score 4 marks should look like:

quantity	does not affect resistance	affects resistance
cross-sectional area		✓
length		✓
volume		✓
mass	✓	

- Cross-sectional area does affect resistance; [1 mark]
- Length does affect resistance; [1 mark]
- Resistivity does affect resistance; [1 mark]
- Mass does not affect resistance; [1 mark]

[Total: 4 marks]

You can determine which quantities the resistance depends on, look at the equation for resistance

$$R = \frac{\rho l}{A}.$$

Therefore, the resistance is affected by the resistivity (ρ), the length (l) and the cross-sectional area (A).

3b

Calculate the cross-sectional area of the metal wire:

List the known quantities:

- Diameter of the wire, $d = 66\ \mu m = 66 \times 10^{-6}$ m

Determine the equation for cross-sectional area, A:

- $A = \pi r^2 = \pi\left(\frac{d}{2}\right)^2 = \frac{\pi d^2}{4}$

Substitute in the values:

$\pi(66 \times 10^{-6})^2$

- $A = \frac{\pi(\ldots)}{4}$ [1 mark]
- $A = 3.421 \times 10^{-9} = 3.4 \times 10^{-9}$ m^2 [1 mark]

[Total: 2 marks]

3c

The resistance of the wire is:

List the known quantities:

- Current, I = 0.65 kA = 0.65 × 10^3 A
- Potential difference, V = 240 V

Calculate the resistance, R:

- $R = \frac{V}{I} = \frac{240}{0.65 \times 10^3}$ [1 mark]
- $R = 0.369 = 0.37\,\Omega$ [1 mark]

[Total: 2 marks]

3d

To determine which metal the filament is made from:

List the known quantities:

- Resistance, R = 0.369 Ω (from **(b)**)
- Cross-sectional area, A = 3.421 × 10^{-9} m^2 (from **(a)**)
- Length, l = 23 mm = 23 × 10^{-3} m

Determine the equation for the resistivity of the wire, ρ:

- $R = \frac{\rho l}{A} \Rightarrow \rho = \frac{RA}{l}$

Substitute in the values:

- $\rho = \frac{0.369 \times (0.3421 \times 10^{-9})}{23 \times 10^{-3}}$ [1 mark]

- $\rho = 5.488 \times 10^{-8} = 5.5 \times 10^{-8}\ \Omega\ \text{m}$ [1 mark]

Identify the metal from Table 1.2:

- (Therefore, the metal is) tungsten; [1 mark]

[Total: 3 marks]